Semester II | Theory · 4 Credits | CIE 40 · ESE 60 | 60 Contact Hours

Business Mathematics

Matrices · Probability · Financial Mathematics · Calculus for Business Decisions

COURSE CODE
2001E

PROGRAMME
Diploma in Commerce / Business Administration

TEACHING SCHEME
L:T:P:R — 3:1:0:0

SELF-LEARNING
82.5 hours (total)

COURSE TYPE
Theory

CREDITS
4

Official Source Declaration

📄 **Source:** This handbook is based on the official **Kerala Polytechnic Revision 2026** syllabus for **Course 2001E — Business Mathematics**, Semester II. The syllabus document (PDF) was used as the sole authoritative source for course objectives, outcomes, modules, topics, subtopics, teaching hours, assessment scheme, and learning resources.

⚠️ **Verified Inconsistencies / Notes:**

- The PDF lists **82.5 self-learning hours** across 15 weeks, but the module-wise self-learning activity descriptions in the syllabus are grouped per week. The handbook preserves all activities as listed.
- The CO–PO matrix shows correlation values (e.g., 3, 2.25) but the programme outcomes (PO1–PO11) are not explicitly named in the PDF. The handbook reproduces the matrix as given.
- The series examination pattern (Part A/B/C) and ESE pattern are reproduced exactly from the PDF. Any minor typographic inconsistencies in the source are preserved as official.
- The module hour totals (16+14+16+13 = 59) match the 60 contact hours closely; the handbook uses the official per-module hour breakdown.

Course Dashboard

4

Credits

60

Contact Hours

82.5

Self-Learning Hours

40

CIE Marks

60

ESE Marks

4

Modules

📘 Course Introduction

Business Mathematics is a foundational course designed to equip diploma students with essential mathematical tools required for effective business decision-making. The course integrates mathematical concepts with real-world business applications such as financial analysis, risk assessment, data organization, and economic decision-making. Emphasis is placed on problem-solving, logical reasoning, and interpretation of results in managerial and commercial contexts, ensuring relevance to modern business environments.

📋 Pre-requisite

Basic Mathematics (Secondary Education level). Students should be comfortable with arithmetic operations, basic algebra, and simple equations.

📊 Teaching & Assessment Scheme

Teaching Scheme per Week

L	T	P	R	Self-Learning (SS)	Total Hours/Sem	Credits	CIE	ESE	Total
3	1	0	0	5.5	60	4	40	60	100

Legend: L=Lecture, T=Tutorial, P=Practical, R=Project, SS=Self-Learning, CIE=Continuous Internal Evaluation, ESE=End Semester Examination.

How to Use This Handbook

1

Understand

Read each module topic in depth. Follow the explanations, formulas, and worked examples.

2

Visualise

Use diagrams, tables, and the visual library to grasp concepts quickly.

3

Apply

Solve the module-wise questions and practice with the model paper. Use the calculators to check your work.

4

Revise

Review the formula bank, quick-revision cards, glossary, and common-mistake list before exams.

Course Objectives & Outcomes

🎯 Course Objectives

1. To develop numerical and analytical skills required for business decision-making.
2. To apply mathematical concepts to real-life business and economic situations.
3. To build financial literacy relevant to banking, trade, and industry.
4. To enable students to interpret data and solve business problems logically.

📌 Course Outcomes (CO)

Course Outcomes with Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Explain and apply matrix operations and determinant concepts to organize and solve structured business data problems.	Apply
CO2	Explain and apply basic probability concepts to analyze business situations involving risk and uncertainty.	Apply
CO3	Compute and apply financial mathematics concepts related to interest, profit, loss, time value of money, and basic investment decisions.	Apply
CO4	Apply basic differentiation and integration techniques to solve simple business and economic problems such as marginal analysis and optimization.	Apply

📊 CO–PO Mapping (Course Articulation Matrix)

CO-PO Correlation (Legends: 1-Low, 2-Medium, 3-High, “-” = No Correlation)											
CO	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1	3	3	2	2	—	—	—	2	—	—	2
CO2	3	3	2	2	—	2	—	—	—	2	—
CO3	3	3	3	2	—	2	—	—	—	3	—
CO4	3	3	2	2	—	2	—	—	—	2	—
Total	12	12	9	8	—	6	—	—	—	9	—
Avg	3	3	2.25	2	—	2	—	—	—	2.25	—

Curriculum Coverage Map

Module-wise coverage with hours, CO mapping, and handbook anchor				
Module	**Title**	**L+T Hours**	**CO**	**Handbook Section**
I	Matrices	12 + 4 = 16	CO1	Module 1
II	Probability	10 + 4 = 14	CO2	Module 2
III	Financial Mathematics	12 + 4 = 16	CO3	Module 3
IV	Calculus & Business Applications	11 + 2 = 13	CO4	Module 4
Total		**59 (≈60)**		

Module Roadmap: How the Modules Connect

📊 Module 1: Matrices

Foundation for organizing business data, cost-output analysis, and structured problem-solving.

🎲 Module 2: Probability

Builds on logical reasoning to handle risk, uncertainty, and quality control in business.

💰 Module 3: Financial Mathematics

Applies arithmetic and algebra to banking, investments, EMI, and profit/loss analysis.

📈 Module 4: Calculus

Uses differentiation and integration for marginal analysis, optimization, and economic decision-making.

Flow: Matrices → Probability → Financial Math → Calculus. Each module builds on the previous; all four are examined in the End Semester Examination.

MODULE I

Matrices

🕒 16 hrs | CO1 | Apply

Matrices are the backbone of structured data organization in business. From cost-output tables to inventory management, matrices help you organize, manipulate, and interpret large sets of numerical information efficiently.

Introduction to Matrices & Business Relevance▾

What is a Matrix? A matrix is a rectangular array of numbers, symbols, or expressions arranged in rows and columns. In business, matrices are used to organize data such as sales figures, cost structures, inventory levels, and production schedules.

Business Relevance: Managers use matrices to compare product performance across regions, track input-output relationships, and solve linear programming problems. For example, a retail chain might use a matrix to store monthly sales data for 5 products across 4 cities.

Malayalam support: □□□ matrix □□□□□□ □□□□□□□□□ (rows) □□□□□□□□□ (columns) □□□□□□□□□□□ □□□□□□□□□□ □□□ □□□□□□□□ □□□□□□□□□□. □□□□□□□□□□□, □□□□□□□ □□□□□□□□, □□□□□ □□□□□, □□□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□□□□□□□ matrices □□□□□□□□□□□□□□.

💡 **Key idea:** A matrix is not just a table — it's a mathematical object that can be added, subtracted, multiplied, and transformed to reveal business insights.

Types of Matrices▾

Matrices are classified based on their shape, elements, and properties. Understanding these types is essential for choosing the right operation in business applications.

Types of Matrices

Type	Description	Example
Row Matrix	Has only one row.	[2 5 8]
Column Matrix	Has only one column.	[2; 5; 8]
Square Matrix	Number of rows = number of columns.	2×2 or 3×3
Diagonal Matrix	All non-diagonal elements are zero.	diag(3,7,2)
Scalar Matrix	Diagonal matrix with all diagonal elements equal.	diag(5,5,5)
Identity Matrix	Diagonal matrix with 1s on the diagonal.	I_2 = [[1,0],[0,1]]
Zero Matrix	All elements are zero.	[[0,0],[0,0]]
Symmetric Matrix	$A = A^T$ (transpose equals itself).	[[1,2],[2,3]]
Skew-Symmetric	$A = -A^T$ (diagonal elements are zero).	[[0,2],[-2,0]]

📐 **EXAMPLE**
Given: Matrix A = [[4,0,0],[0,7,0],[0,0,2]]
Type: Diagonal matrix (since non-diagonal elements are zero). It is also a square matrix of order 3×3.

Malayalam support: [illegible]

Determinant of a Square Matrix▼

Definition: The determinant is a scalar value associated with a square matrix. It is denoted as |A| or det(A). Determinants help in solving systems of linear equations, finding the inverse of a matrix, and checking if a matrix is singular (det = 0).

For a 2×2 matrix: If A = [[a,b],[c,d]], then det(A) = ad − bc.

For a 3×3 matrix: det(A) = a(ei − fh) − b(di − fg) + c(dh − eg), using the first-row expansion.

$$\det(A) = a(ei - fh) - b(di - fg) + c(dh - eg)$$

📐 **WORKED EXAMPLE**
Given: A = [[2,3],[5,7]]
Required: det(A)
Calculation: det(A) = (2×7) − (3×5) = 14 − 15 = **−1**

📐 **3×3 EXAMPLE**
Given: A = [[1,2,3],[0,4,5],[1,0,6]]
Calculation: det = 1(4×6 − 5×0) − 2(0×6 − 5×1) + 3(0×0 − 4×1) = 1(24) − 2(−5) + 3(−4) = 24 + 10 − 12 = **22**

💡 **Business relevance:** In business, determinants can be used to check if a system of equations (e.g., supply-demand equilibrium) has a unique solution. If det = 0, the system may have no solution or infinitely many solutions — a sign of potential instability.

Properties of Determinants▾

Determinants have several useful properties that simplify calculations:

- **Property 1:** The determinant of a matrix and its transpose are equal: $|A| = |A^T|$.
- **Property 2:** If any two rows (or columns) are interchanged, the determinant changes sign.
- **Property 3:** If any row (or column) is multiplied by a scalar k, the determinant is multiplied by k.
- **Property 4:** If any two rows (or columns) are identical, the determinant is zero.
- **Property 5:** Adding a multiple of one row to another does not change the determinant.

📐 **EXAMPLE**
Given: A = [[2,4],[1,3]], det(A) = 2×3 − 4×1 = 2.
Row interchange: Swap rows → [[1,3],[2,4]], det = 1×4 − 3×2 = 4 − 6 = −2 (sign changed).
Row multiplication: Multiply row 1 by 3 → [[6,12],[1,3]], det = 6×3 − 12×1 = 18 − 12 = 6 = 3×2 (multiplied by 3).

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Problems on Determinant Properties▾

Practice is key to mastering determinant properties. Here is a step-by-step problem:

📐 **PROBLEM**

Given: A = [[3,2,1],[1,2,3],[2,1,3]], find det(A) using properties.
Solution: Using row operations: $R2 \leftarrow R2 - (1/3)R1$, $R3 \leftarrow R3 - (2/3)R1$, then expand.
Answer: det(A) = 3(2×3 − 3×1) − 2(1×3 − 3×2) + 1(1×1 − 2×2) = 3(6−3) − 2(3−6) + 1(1−4) = 9 + 6 − 3 = **12**.

💡 **Exam tip:** In exams, you may be asked to find determinants using properties rather than direct expansion. Practice row/column operations to save time.

Matrix Addition▾

Rule: Two matrices of the same order can be added by adding their corresponding elements.

If $A = [[a_{11}, a_{12}],[a_{21}, a_{22}]]$ and $B = [[b_{11}, b_{12}],[b_{21}, b_{22}]]$, then $A + B = [[a_{11}+b_{11}, a_{12}+b_{12}],[a_{21}+b_{21}, a_{22}+b_{22}]]$.

📐 **EXAMPLE**

Given: A = [[2,5],[3,7]], B = [[1,4],[6,2]]
Calculation: A + B = [[2+1, 5+4],[3+6, 7+2]] = [[3,9],[9,9]]

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Matrix Subtraction & Scalar Multiplication▾

Subtraction: $A - B = [[a_{11}-b_{11}, a_{12}-b_{12}],[a_{21}-b_{21}, a_{22}-b_{22}]]$.

Scalar Multiplication: $kA = [[k \cdot a_{11}, k \cdot a_{12}],[k \cdot a_{21}, k \cdot a_{22}]]$, where k is a scalar (real number).

📐 **EXAMPLE**

Given: A = [[4,6],[2,8]], B = [[1,3],[5,2]], k = 3.
Subtraction: A − B = [[3,3],[−3,6]]
Scalar: 3A = [[12,18],[6,24]]

💡 **Business use:** Scalar multiplication can represent scaling of inventory levels or cost adjustments by a constant factor (e.g., 10% price increase).

Conformability for Matrix Multiplication▾

Rule: Two matrices A ($m \times n$) and B ($p \times q$) are conformable for multiplication if the number of columns in A equals the number of rows in B, i.e., $n = p$. The resulting matrix has order $m \times q$.

Example: A (2×3) × B (3×4) → Result (2×4). If $n \neq p$, multiplication is not defined.

> 📐 **EXAMPLE**
> **Given:** A (2×3), B (3×2). Conformable? Yes ($3 = 3$). Result: 2×2.
> **Given:** A (2×3), B (2×3). Conformable? No ($3 \neq 2$).

> **Malayalam support:** [illegible]

Matrix Multiplication▾

Procedure: If A ($m \times n$) and B ($n \times p$), the product $C = AB$ is defined as:

$C_{ij} = \Sigma_{k=1}^{n} A_{ik} \cdot B_{kj}$

Each element C_{ij} is the dot product of the i-th row of A and the j-th column of B.

> 📐 **EXAMPLE**
> **Given:** A = [[2,3],[1,4]], B = [[5,6],[7,8]]
> **Calculation:**
> $C_{11} = 2 \times 5 + 3 \times 7 = 10 + 21 = 31$
> $C_{12} = 2 \times 6 + 3 \times 8 = 12 + 24 = 36$
> $C_{21} = 1 \times 5 + 4 \times 7 = 5 + 28 = 33$
> $C_{22} = 1 \times 6 + 4 \times 8 = 6 + 32 = 38$
> **Result:** AB = [[31,36],[33,38]]

> ⚠️ **Important:** Matrix multiplication is *not* commutative: $AB \neq BA$ in general. Always check conformability first.

Transpose of a Matrix▾

Definition: The transpose of a matrix A, denoted A^T, is obtained by interchanging its rows and columns. If A is $m \times n$, A^T is $n \times m$.

Example: A = [[1,2,3],[4,5,6]] → A^T = [[1,4],[2,5],[3,6]].

📐 **EXAMPLE**

Given: A = [[7,8],[9,10]]
Transpose: A^T = [[7,9],[8,10]]

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Properties of Transpose▾

- **Property 1:** $(A^T)^T = A$ (transposing twice returns the original matrix).
- **Property 2:** $(A + B)^T = A^T + B^T$.
- **Property 3:** $(kA)^T = kA^T$.
- **Property 4:** $(AB)^T = B^T A^T$ (the product reversed).

📐 **EXAMPLE**

Given: A = [[1,2],[3,4]], B = [[5,6],[7,8]].
Verify: $(A+B)^T$ = [[6,10],[8,12]]; A^T+B^T = [[1,3],[2,4]]+[[5,7],[6,8]] = [[6,10],[8,12]]. ✓

Numerical Problems on Transpose▾

📐 **PROBLEM**

Given: A = [[2,1],[−1,0]], B = [[3,4],[5,6]]. Find $(AB)^T$ and verify it equals B^TA^T.
Solution: AB = [[2×3+1×5, 2×4+1×6],[−1×3+0×5, −1×4+0×6]] = [[11,14],[−3,−4]].
$(AB)^T$ = [[11,−3],[14,−4]].
B^TA^T = [[3,5],[4,6]] × [[2,−1],[1,0]] = [[3×2+5×1, 3×−1+5×0],[4×2+6×1, 4×−1+6×0]] = [[11,−3],[14,−4]]. ✓

Business Applications of Matrices (Case-based Practice)▾

Case: Cost-Output Analysis

A factory produces three products: P_1, P_2, P_3. The output matrix (units) over two months is:

Output = [[120, 150, 90], [110, 140, 100]] (rows = months, columns = products).

The cost per unit (₹) is: Cost = [50, 40, 60] (as a row vector).

Find the total cost per month.

📐 **SOLUTION**

Total cost = Output × $Cost^T$ (conformable: 2×3 × 3×1 → 2×1).
Month 1: 120×50 + 150×40 + 90×60 = 6000 + 6000 + 5400 = **₹17,400**.
Month 2: 110×50 + 140×40 + 100×60 = 5500 + 5600 + 6000 = **₹17,100**.

Other applications: Inventory management, supply chain optimization, budgeting, and sales forecasting all rely on matrix operations.

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📌 **Module 1 Summary:** You learned about matrices — types, determinants, properties, operations (addition, subtraction, scalar multiplication, multiplication), transpose, and business applications. Practice determinant problems and matrix multiplication to build speed and accuracy.

MODULE II

Probability

🕒 14 hrs CO2 Apply

Probability is the language of uncertainty. In business, every decision involves risk — from launching a new product to approving a loan. This module gives you the tools to quantify and manage that uncertainty.

Random Experiment & Sample Space▾

Random Experiment: An experiment whose outcome cannot be predicted with certainty, but the set of all possible outcomes is known. Examples: tossing a coin, rolling a die, selecting a customer at random.

Sample Space (S): The set of all possible outcomes of a random experiment. For a coin toss, S = {H, T}. For a die roll, S = {1, 2, 3, 4, 5, 6}.

Event: A subset of the sample space. For example, getting an even number on a die: E = {2, 4, 6}.

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Events and Types of Events▾

Types of Events:

- **Simple Event:** A single outcome (e.g., getting a 3 on a die).
- **Compound Event:** More than one outcome (e.g., getting an even number).
- **Null Event (∅):** An event with no outcomes.
- **Sure Event (S):** The entire sample space.
- **Complementary Event (E'):** All outcomes not in E.

> 📐 **EXAMPLE**
> Die roll: S = {1,2,3,4,5,6}.
> E = {2,4,6} (even numbers). Complement E' = {1,3,5} (odd numbers).

Mutually Exclusive Events▾

Definition: Two events are mutually exclusive if they cannot occur at the same time. That is, their intersection is empty: $A \cap B = \varnothing$.

Example: Tossing a coin: Event A = Heads, Event B = Tails. They are mutually exclusive.

Addition Rule: If A and B are mutually exclusive, $P(A \cup B) = P(A) + P(B)$.

> 📐 **EXAMPLE**
> Die roll: A = {1,2}, B = {5,6}. A and B are mutually exclusive. P(A)=2/6, P(B)=2/6, P(A∪B)=4/6.

Equally Likely Events▾

Definition: Events are equally likely if they have the same probability of occurring.

Example: In a fair die, each face {1,2,3,4,5,6} is equally likely with probability 1/6.

Formula: If all outcomes are equally likely, P(E) = (Number of outcomes in E) / (Total outcomes in S).

> 📐 **EXAMPLE**
> A bag has 3 red, 5 blue, 2 green balls. Probability of drawing a blue ball = 5/(3+5+2) = 5/10 = 0.5.

Exhaustive Events▾

Definition: A set of events is exhaustive if their union covers the entire sample space. That is, at least one of them must occur.

Example: For a die, events $A = \{1,2\}$, $B = \{3,4\}$, $C = \{5,6\}$ are exhaustive because $A \cup B \cup C = S$.

Note: Exhaustive events need not be mutually exclusive.

> **Malayalam support:** □□□ □□□□□□ □□□□□□□□□ □□□□□□□ □□□□□□□ □□□□□□□ □□□□□□□□ □□□□□□□□ □□□□□□□□□□□□□□□□□□□□□□□□ □□ □□□□□□□□□ (exhaustive) □□□□□□□□□ □□□□□□□□□□□□□□□□□□□.

Independent Events▾

Definition: Two events are independent if the occurrence of one does not affect the probability of the other. Mathematically, $P(A \cap B) = P(A) \times P(B)$.

Example: Tossing a coin twice. The outcome of the first toss does not affect the second toss.

> 📐 **EXAMPLE**
>
> $P(A) = 0.4$ (heads on first toss), $P(B) = 0.6$ (heads on second). If independent, $P(A \cap B) = 0.4 \times 0.6 = 0.24$.

Dependent Events▾

Definition: Two events are dependent if the occurrence of one affects the probability of the other. The conditional probability is used: $P(A|B) = P(A \cap B) / P(B)$.

Example: Drawing two cards without replacement from a deck. The probability of the second card being an ace depends on the first card.

> 📐 **EXAMPLE**
>
> A box has 5 red and 5 blue balls. Draw two balls without replacement. $P(\text{second is red} \mid \text{first is red}) = 4/9$.

> 💡 **Business context:** Dependent events appear in inventory management — if a product is out of stock (first event), the probability of a lost sale (second event) changes.

Mixed Numerical Problems▾

📐 **PROBLEM 1**

Given: A and B are mutually exclusive with P(A)=0.3, P(B)=0.5. Find P(A∪B).
Answer: P(A∪B) = 0.3 + 0.5 = 0.8.

📐 **PROBLEM 2**

Given: P(A)=0.6, P(B)=0.4, and A and B are independent. Find P(A∩B).
Answer: P(A∩B) = 0.6 × 0.4 = 0.24.

📐 **PROBLEM 3**

Given: In a class, 60% are boys, 40% are girls. 80% of boys and 70% of girls passed the exam. Find the probability that a randomly chosen student passed.
Solution: P(pass) = P(boy)×P(pass|boy) + P(girl)×P(pass|girl) = 0.6×0.8 + 0.4×0.7 = 0.48 + 0.28 = 0.76.

Probability in Quality Control▾

Application: In manufacturing, probability is used to monitor product quality. For example, if a machine produces 2% defective items, the probability of finding 0 defects in a sample of 10 can be calculated using the binomial distribution.

📐 **EXAMPLE**

A factory produces items with a 3% defect rate. What is the probability that a randomly selected item is not defective?
Answer: P(not defective) = 1 − 0.03 = 0.97.

Quality control charts use probability to set upper and lower control limits for production processes.

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Risk and Uncertainty in Business▾

Risk: Measurable uncertainty where probabilities are known (e.g., insurance).

Uncertainty: Unmeasurable — probabilities are unknown or subjective.

Business example: A company launching a new product estimates a 70% chance of success based on market research. This is risk. If no data exists, it's uncertainty.

💡 **Key distinction:** Risk can be modeled with probability; uncertainty requires judgment and scenario analysis.

Probability in Business Decisions▾

Decision-making under risk: Managers use expected value = Σ (outcome × probability).

📐 **EXAMPLE**

A project has two possible outcomes: Profit ₹10 lakh (probability 0.6) or loss ₹2 lakh (probability 0.4).

Expected value: $10 \times 0.6 + (-2) \times 0.4 = 6 - 0.8 =$ **₹5.2 lakh**.

This helps compare projects, prioritize investments, and set contingency plans.

📌 **Module 2 Summary:** You learned about probability — random experiments, sample space, events (mutually exclusive, equally likely, exhaustive, independent, dependent), and key business applications in quality control, risk, and decision-making. Practice conditional probability and expected value problems.

MODULE III

Financial Mathematics

🕒 16 hrs | CO3 | Apply

Financial mathematics equips you with the skills to handle money — from calculating interest on loans to evaluating investment opportunities. This module is directly applicable to banking, trade, and personal finance.

Introduction to Financial Mathematics▾

What is Financial Mathematics? It is the application of mathematical methods to financial problems — interest calculations, profit/loss analysis, investment evaluation, and time value of money.

Why it matters: Every business decision involving money — borrowing, lending, investing, pricing — requires financial math. Banks, insurance companies, and stock markets all rely on these calculations.

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Simple Interest▾

Formula: SI = (P × R × T) / 100, where P = Principal, R = Rate of interest per annum (%), T = Time in years.

Amount: A = P + SI.

$$SI = (P \times R \times T) / 100$$

📐 **EXAMPLE**

Given: P = ₹50,000, R = 8% p.a., T = 3 years.
SI: (50000 × 8 × 3) / 100 = 500 × 8 × 3 = ₹12,000.
Amount: 50,000 + 12,000 = ₹62,000.

💡 **Use:** Simple interest is used for short-term loans, fixed deposits, and bonds.

Compound Interest▾

Formula: $A = P (1 + r/n)^{nt}$, where r = annual interest rate (decimal), n = number of times interest compounded per year, t = number of years.

Compound Interest: CI = A − P.

$$A = P\,(1 + r/n)^{nt}$$

📐 **EXAMPLE**

Given: P = ₹10,000, r = 10% (0.10), n = 1 (annually), t = 2 years.
A: $10000 \times (1 + 0.10/1)^{1\times2} = 10000 \times (1.10)^2 = 10000 \times 1.21$ = ₹12,100.
CI: 12,100 − 10,000 = ₹2,100.

📐 **EXAMPLE (COMPOUNDED QUARTERLY)**

Given: P = ₹10,000, r = 10%, n = 4, t = 2.
A: $10000 \times (1 + 0.10/4)^{4\times2} = 10000 \times (1.025)^8 \approx$ ₹12,184.03.
CI: ≈ ₹2,184.03 (higher than annual compounding).

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Compound Interest — Solving Problems▾

📐 **PROBLEM**

Given: Find the compound interest on ₹8,000 at 12% p.a. for 1.5 years, compounded half-yearly.

Solution: $r = 0.12$, $n = 2$, $t = 1.5$, $nt = 3$.

$A = 8000 \times (1 + 0.12/2)^3 = 8000 \times (1.06)^3 = 8000 \times 1.191016 = ₹9,528.13$.

CI: 9,528.13 − 8,000 = ₹1,528.13.

💡 **Exam tip:** Pay attention to compounding frequency (annual, half-yearly, quarterly). The more frequent the compounding, the higher the effective interest.

Time Value of Money▾

Concept: A rupee today is worth more than a rupee tomorrow because of its earning potential (interest). This is the core principle of finance.

Future Value (FV): $FV = PV \times (1 + r)^n$.

Present Value (PV): $PV = FV / (1 + r)^n$.

$$FV = PV \times (1 + r)^n \mid PV = FV / (1 + r)^n$$

📐 **EXAMPLE**

Given: PV = ₹5,000, r = 8%, n = 5 years. Find FV.

FV: $5000 \times (1.08)^5 = 5000 \times 1.469328 \approx ₹7,346.64$.

📐 **EXAMPLE (PRESENT VALUE)**

Given: You need ₹10,000 after 3 years. r = 10%. What is the PV?

PV: $10000 / (1.10)^3 = 10000 / 1.331 \approx ₹7,513.15$.

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Percentage Calculations▾

Definition: Percentage means "per 100." It is a dimensionless ratio used to express proportions.

Basic formulas:

- Percentage = (Part / Whole) × 100
- Part = (Percentage × Whole) / 100
- Whole = (Part × 100) / Percentage

> 📐 **EXAMPLE**
> A company has 200 employees. 60% are male. How many males?
> **Calculation:** 200 × (60/100) = 120 males.

> 📐 **EXAMPLE**
> A product price increased from ₹500 to ₹600. What is the percentage increase?
> **Increase:** 600 − 500 = 100. Percentage = (100/500) × 100 = 20%.

Profit and Loss▾

Key terms:

- **Cost Price (CP):** The price at which an item is bought.
- **Selling Price (SP):** The price at which it is sold.
- **Profit:** SP > CP; Profit = SP − CP.
- **Loss:** CP > SP; Loss = CP − SP.
- **Profit %:** (Profit / CP) × 100.
- **Loss %:** (Loss / CP) × 100.

$$\text{Profit} = \text{SP} - \text{CP} \ \mid\ \text{Loss} = \text{CP} - \text{SP}$$

> 📐 **EXAMPLE**
> **Given:** CP = ₹1,200, SP = ₹1,500.
> **Profit:** 1,500 − 1,200 = ₹300.
> **Profit %:** (300/1200) × 100 = 25%.

> 📐 **EXAMPLE**
> **Given:** CP = ₹800, SP = ₹720.
> **Loss:** 800 − 720 = ₹80.
> **Loss %:** (80/800) × 100 = 10%.

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Cost Price & Selling Price Relations▾

Finding SP from CP and profit %: SP = CP × (1 + profit%/100).

Finding SP from CP and loss %: SP = CP × (1 − loss%/100).

Finding CP from SP and profit %: CP = SP × 100 / (100 + profit%).

Finding CP from SP and loss %: CP = SP × 100 / (100 − loss%).

📐 **EXAMPLE**

Given: CP = ₹2,000, profit % = 15%. Find SP.
SP: 2000 × (1 + 15/100) = 2000 × 1.15 = ₹2,300.

📐 **EXAMPLE**

Given: SP = ₹1,250, loss % = 10%. Find CP.
CP: 1250 × 100 / (100 − 10) = 125000 / 90 = ₹1,388.89.

💡 **Business use:** Retailers use these formulas to set prices, evaluate discounts, and manage margins.

EMI (Equated Monthly Installment)▾

Definition: EMI is a fixed payment amount made by a borrower to a lender each month, covering both interest and principal.

Formula: $EMI = [P \times r \times (1+r)^n] / [(1+r)^n - 1]$, where P = loan amount, r = monthly interest rate (annual rate / 12), n = number of monthly installments.

$$EMI = [P \times r \times (1+r)^n] / [(1+r)^n - 1]$$

📐 **EXAMPLE**

Given: Loan = ₹1,00,000, annual rate = 12% ($r = 0.01$ monthly), tenure = 12 months.

EMI: $[100000 \times 0.01 \times (1.01)^{12}] / [(1.01)^{12} - 1]$

$(1.01)^{12} \approx 1.126825$. Numerator = $100000 \times 0.01 \times 1.126825 = 1{,}126.825$.
Denominator = 0.126825.
EMI ≈ 1,126.825 / 0.126825 ≈ **₹8,884.88**.

💡 **Real-world:** EMI calculators are used by banks and NBFCs for auto loans, home loans, and personal loans.

Investment Comparison (Case-based Learning)▼

Case: You have ₹50,000 to invest. Option A: FD at 8% simple interest for 3 years. Option B: Mutual fund with expected return of 10% compounded annually for 3 years. Which is better?

📐 **SOLUTION**

Option A (SI): SI = $(50000 \times 8 \times 3)/100$ = ₹12,000. Total = ₹62,000.
Option B (CI): $A = 50000 \times (1.10)^3 = 50000 \times 1.331$ = ₹66,550.
Difference: ₹4,550 more in Option B. But consider risk — FD is safer, mutual fund carries market risk.

Investment decisions always balance return and risk. Financial math helps quantify the return side of the equation.

Malayalam support: [illegible]

Inflation (Introductory)▼

Definition: Inflation is the rate at which the general level of prices for goods and services rises, eroding purchasing power.

Impact on money: If inflation is 6% p.a., ₹100 today will be worth only ₹94.34 in one year in terms of purchasing power (100 / 1.06).

Real rate of return: Real rate = Nominal rate − Inflation rate.

📐 **EXAMPLE**

A bank offers 8% interest. Inflation is 5%. Real return = 8% − 5% = 3%.

> 💡 **Why it matters:** Inflation reduces the real value of future cash flows. Businesses must consider inflation when pricing, budgeting, and forecasting.

> 📌 **Module 3 Summary:** You learned about simple and compound interest, time value of money, percentages, profit/loss, EMI, and investment comparison. These tools are essential for banking, trade, and personal financial planning.

MODULE IV

Calculus & Business Applications

🕒 13 hrs **CO4** **Apply**

Calculus gives you the power to analyze change — how a cost function behaves as production increases, how revenue changes with price, and how to find the optimal point where profit is maximized. This module bridges mathematics and economic decision-making.

Introduction to Calculus in Business▾

Why calculus in business? Business problems often involve rates of change: revenue growth, cost increase, profit optimization. Calculus provides the language and tools to model these changes precisely.

Two main branches:

- **Differential calculus:** Studies rates of change (derivatives).
- **Integral calculus:** Studies accumulation (areas, totals).

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Differentiation from First Principle▾

Definition: The derivative of a function f(x) at a point x is the limit of the average rate of change as the interval approaches zero:

$$f'(x) = \lim_{h \to 0} [f(x+h) - f(x)] / h$$

This is the "first principle" or "delta method."

📐 **EXAMPLE**

Given: $f(x) = x^2$. Find $f'(x)$ from first principle.

$f'(x) = \lim_{h\to0} [(x+h)^2 - x^2] / h$

$= \lim [(x^2 + 2xh + h^2) - x^2] / h$

$= \lim [2xh + h^2] / h = \lim (2x + h) = 2x$.

💡 **Note:** In practice, we use standard derivative rules instead of the first principle, but understanding the principle is essential for conceptual clarity.

Derivatives of Standard Functions▾

$$\frac{d}{dx}(x^n) = n\,x^{n-1}$$

Power rule

$$\frac{d}{dx}(e^x) = e^x$$

Exponential

$$\frac{d}{dx}(\ln x) = 1/x$$

Natural log

$$\frac{d}{dx}(\sin x) = \cos x$$

Sine

$$\frac{d}{dx}(\cos x) = -\sin x$$

Cosine

$$\frac{d}{dx}(k) = 0$$

Constant

📐 **EXAMPLE**

Given: $f(x) = 3x^4 + 2x - 5$. Find $f'(x)$.

Solution: $f'(x) = 3(4x^3) + 2(1) - 0 = 12x^3 + 2$.

Rules of Differentiation▾

- **Constant rule:** $d/dx\,(c) = 0$.

- **Power rule:** $d/dx\,(x^n) = n\,x^{n-1}$.
- **Sum rule:** $d/dx\,(u+v) = u' + v'$.
- **Difference rule:** $d/dx\,(u-v) = u' - v'$.
- **Product rule:** $d/dx\,(uv) = u'v + uv'$.
- **Quotient rule:** $d/dx\,(u/v) = (u'v - uv') / v^2$.
- **Chain rule:** $d/dx\, f(g(x)) = f'(g(x)) \cdot g'(x)$.

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Product Rule▾

$$(uv)' = u'v + uv'$$

📐 **EXAMPLE**

Given: $f(x) = x^2 \cdot \sin x$. Find $f'(x)$.

Solution: $u = x^2$, $u' = 2x$; $v = \sin x$, $v' = \cos x$.

$f'(x) = 2x \cdot \sin x + x^2 \cdot \cos x$.

Quotient Rule▾

$$(u/v)' = (u'v - uv') / v^2$$

📐 **EXAMPLE**

Given: $f(x) = (x^2 + 1) / (x - 1)$. Find $f'(x)$.

Solution: $u = x^2+1$, $u' = 2x$; $v = x-1$, $v' = 1$.

$f'(x) = [2x(x-1) - (x^2+1)(1)] / (x-1)^2 = (2x^2-2x - x^2-1) / (x-1)^2 = (x^2-2x-1) / (x-1)^2$.

Mixed Differentiation Problems▾

📐 **PROBLEM 1**

Given: $f(x) = (2x^3 + 3x)(x^2 - 1)$. Find $f'(x)$.

Solution: Use product rule: $u = 2x^3+3x$, $u' = 6x^2+3$; $v = x^2-1$, $v' = 2x$.
$f' = (6x^2+3)(x^2-1) + (2x^3+3x)(2x)$.

📐 **PROBLEM 2**

Given: $f(x) = (x^2+2x) / (x+1)$. Find $f'(x)$.

Solution: Use quotient rule: $u = x^2+2x$, $u' = 2x+2$; $v = x+1$, $v' = 1$.
$f' = [(2x+2)(x+1) - (x^2+2x)(1)] / (x+1)^2 = (2x^2+4x+2 - x^2-2x) / (x+1)^2 = (x^2+2x+2) / (x+1)^2$.

Introduction to Integration▾

Definition: Integration is the reverse process of differentiation. It finds the function whose derivative is given. It also represents the area under a curve.

Indefinite integral: $\int f(x)\,dx = F(x) + C$, where $F'(x) = f(x)$ and C is the constant of integration.

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Integrals of Standard Forms▾

$$\int x^n\,dx = x^{n+1}/(n+1) + C$$

$n \neq -1$

$$\int e^x\,dx = e^x + C$$

Exponential

$$\int (1/x)\,dx = \ln|x| + C$$

Natural log

$$\int \sin x\,dx = -\cos x + C$$

Sine

$$\int \cos x\,dx = \sin x + C$$

Cosine

$\int k\,dx = kx + C$

Constant

📐 **EXAMPLE**

Given: $\int (3x^2 + 2x)\,dx$.

Solution: $3\int x^2\,dx + 2\int x\,dx = 3(x^3/3) + 2(x^2/2) + C = x^3 + x^2 + C$.

Integration of Products & Quotients▾

Integration by parts: $\int u\,dv = uv - \int v\,du$.

Integration of quotients: Often requires substitution or partial fractions (beyond this scope).

📐 **EXAMPLE (BY PARTS)**

Given: $\int x e^x\,dx$.

Let $u = x$, $dv = e^x\,dx \rightarrow du = dx$, $v = e^x$.

$\int x e^x\,dx = x e^x - \int e^x\,dx = x e^x - e^x + C$.

💡 **Note:** In business calculus, you typically work with polynomials and simple exponentials — integration by parts is useful for more complex functions.

Business Meaning of Derivatives (Marginal Concepts)▾

Marginal Cost: MC = d(TC)/dQ — the cost of producing one additional unit.

Marginal Revenue: MR = d(TR)/dQ — the revenue from selling one additional unit.

Marginal Profit: MP = MR − MC.

📐 **EXAMPLE**

Given: Total cost $TC(Q) = 0.5Q^2 + 20Q + 100$. Find marginal cost at $Q = 50$.

MC: $d(TC)/dQ = Q + 20$. At Q=50, MC = 50 + 20 = ₹70 per unit.

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Profit Maximization (Case-based)▾

Rule: Profit is maximized where Marginal Revenue = Marginal Cost (MR = MC). At this point, the slope of the profit function is zero.

📐 **EXAMPLE**

Given: $TR(Q) = 100Q - 0.5Q^2$, $TC(Q) = 0.2Q^2 + 10Q + 50$.
Find profit-maximizing Q.
$MR = d(TR)/dQ = 100 - Q$.
$MC = d(TC)/dQ = 0.4Q + 10$.
Set MR = MC: $100 - Q = 0.4Q + 10 \rightarrow 90 = 1.4Q \rightarrow Q \approx 64.29$.
Maximum profit at $Q \approx 64$ units.

💡 **Business insight:** This is how firms decide production levels — they produce until the cost of the last unit equals the revenue it brings in.

Comprehensive Application Problems▾

📐 **PROBLEM**

Given: A company's revenue function is $R(x) = 200x - 2x^2$ and cost function is $C(x) = 0.5x^2 + 80x + 200$. Find the profit function, marginal profit, and the output level that maximizes profit.
Profit $P(x) = R(x) - C(x)$: $(200x - 2x^2) - (0.5x^2 + 80x + 200) = 120x - 2.5x^2 - 200$.
Marginal profit: $P'(x) = 120 - 5x$.
Set $P'(x)=0$: $120 - 5x = 0 \rightarrow x = 24$.
Max profit at $x = 24$ units. $P(24) = 120(24) - 2.5(576) - 200 = 2880 - 1440 - 200 = ₹1,240$.

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📌 **Module 4 Summary:** You learned about differentiation (from first principle, standard derivatives, product/quotient rules) and integration (standard forms, by parts). You applied these to marginal analysis and profit maximization — core tools for economic decision-making.

Formula Bank

Quick reference of all important formulas from the four modules.

$$\det(A) = ad - bc$$
2×2 determinant
M1

$$\det(A) = a(ei-fh) - b(di-fg) + c(dh-eg)$$
3×3 determinant
M1

$$(AB)^T = B^T A^T$$
Transpose of product
M1

$$P(E) = n(E) / n(S)$$
Probability (equally likely)
M2

$$P(A\cup B) = P(A) + P(B) - P(A\cap B)$$
Addition rule (non-mutually exclusive)
M2

$$P(A\cap B) = P(A) \times P(B)$$
Independent events
M2

$$P(A|B) = P(A\cap B) / P(B)$$
Conditional probability
M2

$$SI = (P\times R\times T) / 100$$
Simple interest
M3

$$A = P(1 + r/n)^{nt}$$
Compound interest (amount)
M3

$$FV = PV(1+r)^n$$
Future value
M3

$$PV = FV/(1+r)^n$$
Present value
M3

$$\text{Profit}\% = (\text{Profit}/CP)\times 100$$
Profit percentage
M3

$$\text{Loss}\% = (\text{Loss}/CP)\times 100$$
Loss percentage
M3

$$EMI = [P\times r\times(1+r)^n]/[(1+r)^n-1]$$
Equated Monthly Installment
M3

$$d/dx\,(x^n) = n\,x^{n-1}$$
Power rule
M4

$$(uv)' = u'v + uv'$$
Product rule
M4

$$(u/v)' = (u'v - uv')/v^2$$

Quotient rule

M4

$$\int x^n\,dx = x^{n+1}/(n+1) + C$$

Power rule (integration)

M4

$$\int e^x\,dx = e^x + C$$

Exponential integral

M4

$$MC = d(TC)/dQ$$

Marginal cost

M4

$$MR = d(TR)/dQ$$

Marginal revenue

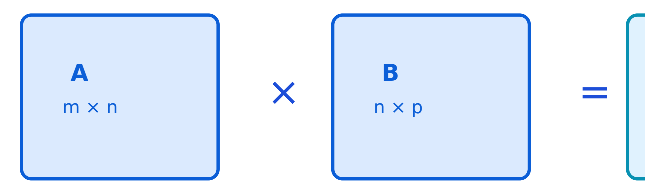
M4

$$\text{Profit max: } MR = MC$$

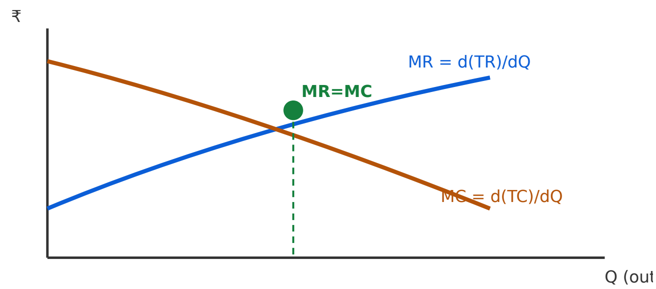
Optimal output condition

M4

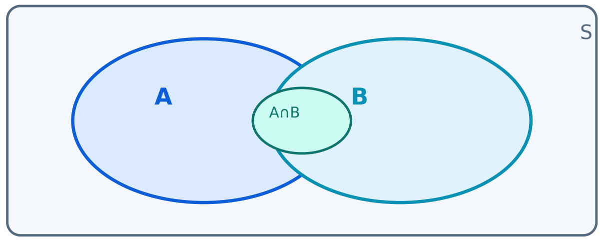
Diagram Library for Rapid Revision



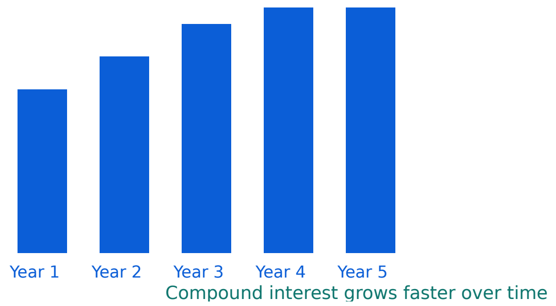
Matrix multiplication conformability: A (m×n) × B (n×p) = C (m×p). The inner dimensions (n) must match.



Profit maximization: The intersection of MR and MC determines the optimal output level.



Sample space S with events A and B. The overlapping region is A∩B.



Compound interest builds faster than simple interest because interest earns interest.

Self-Learning Activities

Official suggested student activities for self-learning (total 82.5 hours).

MODULE 1 · WEEK 1

Concept Map: Types of Matrices

Study the topics taught in class. Prepare a concept map on types of matrices with business examples.

5.5 hrs

MODULE 1 · WEEK 2

Determinant Properties with Spreadsheet

Study class topics. Solve worksheet on determinant properties using spreadsheet tools.

5.5 hrs

MODULE 1 · WEEK 3

Case: Matrix Applications in Cost-Output Analysis

Study class topics. Work on case-based problems on matrix applications in cost-output analysis.

5.5 hrs

MODULE 1 · WEEK 4

Real-life Matrix Problems

Study class topics. Apply matrices to simple real-life business problems.

5.5 hrs

MODULE 2 · WEEK 5

Glossary of Probability Terms

Study class topics. Prepare glossary of probability terms with real-life business examples.

5.5 hrs

MODULE 2 · WEEK 6

Quality Control Probability Problems

Study class topics. Solve probability problems related to quality control scenarios.

5.5 hrs

MODULE 2 · WEEK 7

Mini Assignment: Risk and Uncertainty

Study class topics. Mini assignment on risk and uncertainty in business decisions.

5.5 hrs

MODULE 2 · WEEK 8

Analyze Business Risk Situations

Study class topics. Analyze business risk situations using probability.

5.5 hrs

MODULE 3 · WEEK 9

Simple Interest for Banking Products

Study class topics. Calculate simple interest for different banking products.

5.5 hrs

MODULE 3 · WEEK 10

Compound Interest Comparison

Study class topics. Comparison study of compound interest across banks.

5.5 hrs

MODULE 3 · WEEK 11

Time Value of Money with EMI Calculators

Study class topics. Time value of money problems using EMI calculators.

5.5 hrs

MODULE 3 · WEEK 12

Retail Case Studies: Percentage & Profit-Loss

Study class topics. Percentage and profit-loss calculations from retail case studies.

5.5 hrs

MODULE 4 · WEEK 13

Differentiation Practice

Study class topics. Practice problems on differentiation using standard functions.

5.5 hrs

MODULE 4 · WEEK 14

Marginal Cost & Revenue Case Problems

Study class topics. Business case problems on marginal cost and revenue.

5.5 hrs

MODULE 4 · WEEK 15

Optimization & Reflective Report

Study class topics. Optimization problems for profit maximization. Reflective report on use of mathematics in business decision-making.

5.5 hrs

Assessment Methodology & Examination Pattern

📋 Assessment Scheme

Assessment breakdown

Mode	Details	Duration	Marks	Converted to	Tentative Schedule
CA1	Self-learning activities (Module 1)	—	—	—	End of semester
CA2	Self-learning activities (Module 2)	—	—	—	By 4th week
CA3	Self-learning activities (Module 3)	—	—	—	By 7th week
CA4	Self-learning activities (Module 4)	—	—	—	By 11th week
CA5	Series Exam (2 modules)	2 hours	15	15	By 14th week
CA6	Series Exam (2 modules)	2 hours	15	15	8th week
ESE	Written Exam (theory)	2.5 hours	50	60	15th week
Total CIE (CA1–CA6)			—	**40**	End of semester
Total ESE			50	**60**	End of semester

📝 **SERIES EXAMINATION PATTERN (THEORY)**

Series exam (2 hours)					
Time	**Module**	**Part A**	**Part B**	**Part C (with choice)**	**Total**
2 hours	Any one module	21 marks	3×3 = 9	3×7 = 21	25
	Any one module	21 marks	3×3 = 9	3×7 = 21	25
Total		4×1 = 4	6×3 = 18	4×7 = 28	**50**

📝 **END SEMESTER EXAMINATION PATTERN (THEORY)**

ESE (2.5 hours, 60 marks)					
Duration	**Module**	**Part A (2 Qs × 1 mark)**	**Part B (2 out of 3 Qs × 3 marks)**	**Part C (1 set × 7 marks with choice)**	**Total**
2.5 hours	1	2	2 × 3 = 6	1 set × 7	15
	2	2	2 × 3 = 6	1 set × 7	15
	3	2	2 × 3 = 6	1 set × 7	15
	4	2	2 × 3 = 6	1 set × 7	15
Total		8 × 1 = 8	8 × 3 = 24	4 × 7 = 28	**60**

Module-wise Questions with Answers

Module 1: Matrices

Q1: Define a matrix. List the types of matrices with one example each.▼

Definition: A matrix is a rectangular array of numbers, symbols, or expressions arranged in rows and columns.
Types: Row matrix (e.g., [1 2 3]), Column matrix (e.g., [1;2;3]), Square matrix (2×2), Diagonal matrix (diag(3,7)), Scalar matrix (diag(5,5)), Identity matrix (I_2), Zero matrix ([[0,0],[0,0]]).

Q2: Find the determinant of A = [[2,3,1],[4,5,6],[7,8,9]].▼

Solution: det(A) = 2(5×9 − 6×8) − 3(4×9 − 6×7) + 1(4×8 − 5×7)
= 2(45 − 48) − 3(36 − 42) + 1(32 − 35)

$= 2(-3) - 3(-6) + (-3) = -6 + 18 - 3 = \mathbf{9}$

Q3: Explain the conformability condition for matrix multiplication. Give an example.▼

Condition: Two matrices A (m×n) and B (p×q) are conformable for multiplication if n = p (number of columns of A = number of rows of B).
Example: A (2×3) and B (3×4) are conformable; result is 2×4. If A (2×3) and B (2×3), they are not conformable (3≠2).

Q4: State any four properties of determinants.▼

- $|A| = |A^T|$ (transpose same).
- Interchanging two rows changes sign.
- Multiplying a row by k multiplies determinant by k.
- If two rows are identical, determinant is zero.

Q5: Give a business application of matrices with a brief explanation.▼

Cost-Output Analysis: A matrix can store production quantities (rows = products, columns = months), and a cost vector can be multiplied to find total cost per month. This helps managers budget and plan production efficiently.

Module 2: Probability

Q1: Define sample space and event. Give an example.▼

Sample space (S): The set of all possible outcomes of a random experiment. Example: For a die roll, S = {1,2,3,4,5,6}.
Event: A subset of the sample space. Example: Getting an even number = {2,4,6}.

Q2: What are mutually exclusive events? Give an example.▼

Definition: Two events are mutually exclusive if they cannot occur at the same time ($A \cap B = \varnothing$).
Example: Tossing a coin — Heads and Tails are mutually exclusive.

Q3: Explain independent and dependent events with examples.▼

Independent: Occurrence of one does not affect the other. $P(A \cap B) = P(A) \times P(B)$. Example: Tossing two coins.

Dependent: Occurrence of one affects the probability of the other. Example: Drawing two cards without replacement.

Q4: A bag has 4 red, 5 blue, and 3 green balls. Find the probability of drawing a blue ball.▼

Total balls = 4 + 5 + 3 = 12.
P(blue) = Number of blue balls / Total = 5/12 ≈ **0.4167**.

Q5: Give two business applications of probability.▼

- **Quality control:** Monitoring defect rates in manufacturing.
- **Risk management:** Estimating the probability of loan default or project failure.

Module 3: Financial Mathematics

Q1: Write the formula for simple interest and explain each term.▼

Formula: SI = (P × R × T) / 100
Where: P = Principal (initial amount), R = Rate of interest per annum (%), T = Time in years.

Q2: Find compound interest on ₹20,000 at 10% p.a. for 2 years compounded annually.▼

$A = P(1 + r)^t = 20000 \times (1.10)^2 = 20000 \times 1.21 =$ ₹24,200.
CI = A − P = 24,200 − 20,000 = **₹4,200**.

Q3: Explain the Time Value of Money concept. Give a numerical example.▼

Concept: A rupee today is worth more than a rupee tomorrow because of earning potential (interest).
Example: PV = ₹1,000, r = 8%, n = 3 years. $FV = 1000 \times (1.08)^3 =$ ₹1,259.71. The future value is higher than the present value.

Q4: A product is bought for ₹1,500 and sold for ₹1,800. Find profit percentage.▼

Profit = SP − CP = 1,800 − 1,500 = ₹300.
Profit% = (300/1500) × 100 = **20%**.

Q5: What is EMI? Write the formula and state its business relevance.▼

EMI: Equated Monthly Installment — a fixed monthly payment covering interest and principal.
Formula: $EMI = [P \times r \times (1+r)^n] / [(1+r)^n - 1]$
Relevance: Used in home loans, auto loans, and personal loans to structure repayments.

Module 4: Calculus & Business Applications

Q1: Differentiate $f(x) = 4x^3 - 3x^2 + 5x - 7$.▼

$f'(x) = 4(3x^2) - 3(2x) + 5(1) - 0$
$= \mathbf{12x^2 - 6x + 5}$

Q2: State the product rule and quotient rule of differentiation.▼

Product rule: $(uv)' = u'v + uv'$
Quotient rule: $(u/v)' = (u'v - uv') / v^2$

Q3: Integrate $\int (6x^2 + 4x + 2)\,dx$.▼

$\int 6x^2\,dx = 6(x^3/3) = 2x^3$
$\int 4x\,dx = 4(x^2/2) = 2x^2$
$\int 2\,dx = 2x$
Result: $2x^3 + 2x^2 + 2x + C$

Q4: Explain marginal cost and marginal revenue with a business example.▼

Marginal Cost (MC): The cost of producing one additional unit. $MC = d(TC)/dQ$.
Marginal Revenue (MR): The revenue from selling one additional unit. $MR = d(TR)/dQ$.
Example: A factory's $TC = 0.2Q^2 + 15Q + 200$. $MC = 0.4Q + 15$. At $Q=50$, $MC = 35$. This tells the manager the cost of the 51st unit.

Q5: How do you find the profit-maximizing output level using calculus?▼

Steps:

1. Find profit function: $P(Q) = TR(Q) - TC(Q)$.
2. Differentiate $P(Q)$ to get marginal profit: $P'(Q)$.
3. Set $P'(Q) = 0$ and solve for Q.
4. Verify it's a maximum (second derivative negative).

Alternative: Set MR = MC.

Practice Model Question Paper

📌 **Note:** This is a *practice model paper* based on the official ESE pattern. It is not an official SITTTR model question paper.

Course: 2001E — Business Mathematics | Time: 2.5 hours | Max Marks: 60

PART A — (8 × 1 = 8 MARKS)

Answer all questions. Each carries 1 mark.

1. Define a square matrix.
2. Write the formula for the determinant of a 2×2 matrix.
3. What are mutually exclusive events?
4. If P(A) = 0.4 and P(B) = 0.5, and A and B are independent, find $P(A \cap B)$.
5. Write the formula for simple interest.
6. What is the formula for profit percentage?
7. Find the derivative of x^5.
8. State the marginal revenue condition for profit maximization.

PART B — (8 × 3 = 24 MARKS)

Answer any two questions from each module. Each carries 3 marks.

Module 1:

1. Find the transpose of A = [[2,3,4],[5,6,7]].
2. Verify $(A+B)^T = A^T + B^T$ for A = [[1,2],[3,4]], B = [[5,6],[7,8]].
3. Find the determinant of A = [[1,2],[3,4]].

Module 2:

1. A die is rolled. Find P(even number).
2. Explain exhaustive events with an example.
3. If P(A) = 0.6, P(B) = 0.3, and $P(A \cap B) = 0.2$, find $P(A \cup B)$.

Module 3:

1. Find SI on ₹15,000 at 7% p.a. for 4 years.
2. A product sold for ₹2,400 at a loss of 20%. Find the cost price.
3. Calculate compound interest on ₹8,000 at 6% p.a. for 2 years compounded annually.

Module 4:

1. Differentiate $f(x) = 3x^2 - 2x + 1$.
2. Integrate $\int (2x + 3)\,dx$.
3. Explain marginal cost with a formula.

PART C — (4 × 7 = 28 MARKS)

Answer one set from each module. Each set carries 7 marks (with internal choice).

Module 1: Find the product AB where A = [[2,3],[1,4]] and B = [[5,6],[7,8]]. OR explain the properties of determinants with examples.

Module 2: A box contains 6 red, 4 blue, and 5 green balls. Two balls are drawn without replacement. Find the probability that both are red. OR explain independent and dependent events with examples.

Module 3: A loan of ₹2,00,000 is taken at 10% p.a. for 3 years. Find the EMI (monthly compounding). OR explain time value of money with a numerical example.

Module 4: The revenue function is $R(x) = 50x - 0.5x^2$ and cost function is $C(x) = 0.2x^2 + 10x + 100$. Find the profit-maximizing output and maximum profit. OR differentiate $f(x) = (x^2 + 1)(x - 1)$ using the product rule.

Quick Revision

📊 Module 1: Matrices

- **Types:** Row, Column, Square, Diagonal, Scalar, Identity, Zero, Symmetric
- **Determinant:** 2×2: ad−bc; 3×3: expansion formula
- **Operations:** Addition/Subtraction (same order), Scalar multiplication, Multiplication (conformability: n=p)
- **Transpose:** $(AB)^T = B^T A^T$
- **Business:** Cost-output, inventory, sales data

🎲 Module 2: Probability

- **Sample space:** Set of all outcomes
- **Events:** Simple, Compound, Null, Sure, Complementary
- **Mutually exclusive:** $A \cap B = \varnothing$, $P(A \cup B) = P(A) + P(B)$
- **Independent:** $P(A \cap B) = P(A) \times P(B)$
- **Dependent:** $P(A|B) = P(A \cap B)/P(B)$
- **Business:** Quality control, risk assessment

💰 Module 3: Financial Math

- **SI:** (P×R×T)/100
- **CI:** $A = P(1+r/n)^{nt}$
- **TVM:** $FV = PV(1+r)^n$, $PV = FV/(1+r)^n$
- **Profit/Loss:** Profit%= (Profit/CP)×100
- **EMI:** $[P \times r \times (1+r)^n]/[(1+r)^n - 1]$
- **Inflation:** Real return = Nominal − Inflation

📈 Module 4: Calculus

- **Derivative:** Power rule: $d/dx(x^n)=nx^{n-1}$
- **Product:** (uv)'=u'v+uv'
- **Quotient:** $(u/v)'=(u'v-uv')/v^2$
- **Integration:** $\int x^n\,dx = x^{n+1}/(n+1)+C$
- **Marginal:** MC=d(TC)/dQ, MR=d(TR)/dQ
- **Max profit:** MR = MC

⚠️ **Common Mistakes to Avoid**

- **Matrix multiplication:** Always check conformability (columns of A = rows of B).
- **Determinants:** Don't confuse with matrix — determinant is a scalar.
- **Probability:** Don't assume events are independent unless stated.
- **Compound interest:** Check compounding frequency (annual, half-yearly, quarterly).
- **Profit/Loss:** Profit% and Loss% are always calculated on CP, not SP.
- **Differentiation:** Don't forget the chain rule when functions are composite.
- **Integration:** Always add +C for indefinite integrals.

✅ **Exam Checklist**

- Know all formulas by heart — write them down at the start of the exam.
- Read the question carefully — identify whether it's a definition, application, or numerical.
- Show step-by-step working for numerical problems.
- Check units and percentages — use ₹, %, years correctly.
- Attempt all parts — even if you're unsure, write the formula.
- Manage time: Part A (8 min), Part B (24 min), Part C (28 min).
- Review your answers for calculation errors.

Glossary of Important Terms

Key terms and their meanings

Term	Meaning
Matrix	Rectangular array of numbers arranged in rows and columns.
Determinant	Scalar value associated with a square matrix.
Transpose	Matrix obtained by interchanging rows and columns.
Sample space	Set of all possible outcomes of a random experiment.
Mutually exclusive	Events that cannot occur together.
Independent events	Events where one does not affect the probability of the other.
Conditional probability	Probability of an event given that another event has occurred.
Simple Interest	Interest calculated only on the principal amount.
Compound Interest	Interest calculated on principal plus accumulated interest.
Time Value of Money	Principle that a rupee today is worth more than a rupee tomorrow.
EMI	Equated Monthly Installment — fixed monthly payment for a loan.
Marginal Cost	Cost of producing one additional unit.
Marginal Revenue	Revenue from selling one additional unit.
Profit Maximization	Output level where MR = MC.
Derivative	Rate of change of a function.
Integral	Anti-derivative; area under a curve.

Learning Resources

📚 Textbooks

Prescribed textbooks

Sl. No.	Title	Author	Publication
1	Business Mathematics	Dorai Rajan & Sundaresan	S. Chand
2	Business Mathematics and Statistics	P.R. Vittal	Margham Publications

📖 Reference Books

Additional reference books

Sl. No.	Title	Author	Publication
1	Business Mathematics	Sancheti & Kapoor	S. Chand
2	Mathematics for Management	Raghavachari	Tata McGraw-Hill
3	Business Mathematics	V.K. Kapoor	S. Chand

🌐 Web Resources

Officially listed web resources

Sl. No.	URL	Modules Covered
1	NPTEL: Mathematics for Management	I
2		II
3		General